ETH zürich

Computing Energy Optimisation for D-ITET

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P&S Let’s make ITET green
09/12/2025, ETH Zürich

Introduction

Goals:

- Understand usage of computing resources & energy consumption
- Assess current monitoring gaps & inconsistent power data
- Contribute to analysis of D-ITET computing infrastructure

Methodology:

- Measure power consumption of infrastructure
- Assess and adapt existing scripts
- Analyse and evaluate utilised server components

Resources:

- Zabbix
- ETH Inventory

List of Content

Qualitative Strategy

The Multi-Layered Strategy

The 4 Levels of Measurement:

1. Facility Level
2. Server Level
3. System Level
4. Component Level

Facility Level (PDU / smartplug)

Core Metric: Total Wall-Draw (Watts)

Interface: SNMP queries to Rack PDUs (APC, Eaton) or Smart Plugs.

Scope: Captures the complete energy footprint entering the rack.

Server Level (BMC)

Core Metric: Total Hardware Power (Watts)

Interface: Out-of-Band Management via BMC (Baseboard Management Controller).

Scope: Measures the entire chassis (Motherboard, Disks, Fans, RAM, CPU).

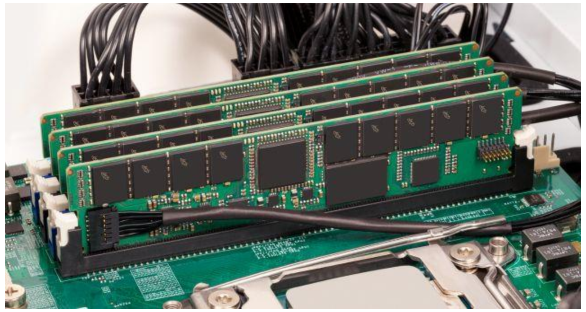
System Level (OS/RAPL)

Core Metric: CPU & DRAM Power (Watts)

Interface: In-Band, OS-level tools (RAPL: **R**unning **A**verage **P**ower **L**imit)

Scope: Active Silicon (CPU, DRAM) — excludes fans, PSUs, and peripherals.

$$P_{\text{watts}} = \frac{E_{t_2} - E_{t_1}}{\Delta t}$$



Component Level (GPU)

Core Metric: Instantaneous GPU Power Draw (Watts)

Interface: Vendor-specific drivers.

- **NVIDIA:** nvidia-smi (NVML)
- **AMD:** rocm-smi

Scope: Dedicated video memory and compute cores.

The Accuracy Challenge (The "Blind Spot")

System (RAPL) and GPU sensors ignore "Base Load."

Unmeasured Components:

Cooling (Fans)

Mainboard Chipsets

PSU (Power Supply Unit) conversion losses & leakage.

Result: Variable error margin inversely proportional to load.

The Accuracy Challenge (The "Blind Spot")

Idle State (0% Load): High Relative Error.

Active silicon is effectively off, but base load (fans/PSU) remains.

Full Load (100% Load): Optimal Correlation.

Processors draw peak power, making the base load a negligible fraction.

Zabbix Monitoring Data

ZABBIX

Datasets

Zabbix Monitoring Data

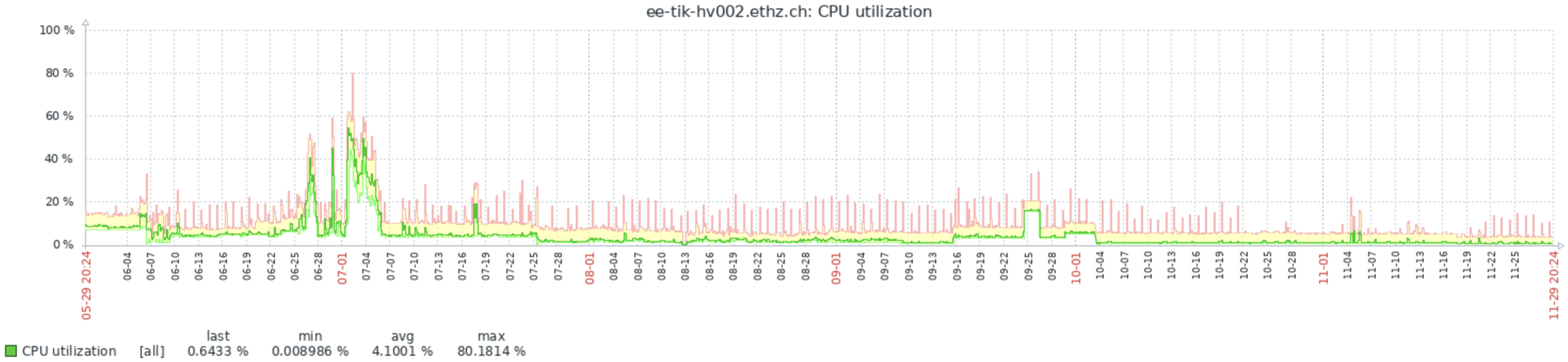
- Open-source monitoring system
- Used to supervise IT/OT infrastructures
- At D-ITET it provides an overview of all monitored hosts, tags, and metrics
- Monitored environment includes
  - Physical compute nodes (ee-tik-cnXXX)
  - Virtual machines (ee-tik-vmXXX)
  - Hypervisors (ee-tik-hvXXX)
  - Workstations
- Provides broad operational data but significant gaps for energy analysis
- Detailed performance metrics exist for almost all hosts
- Power consumption data is available ony for few systems

Datasets

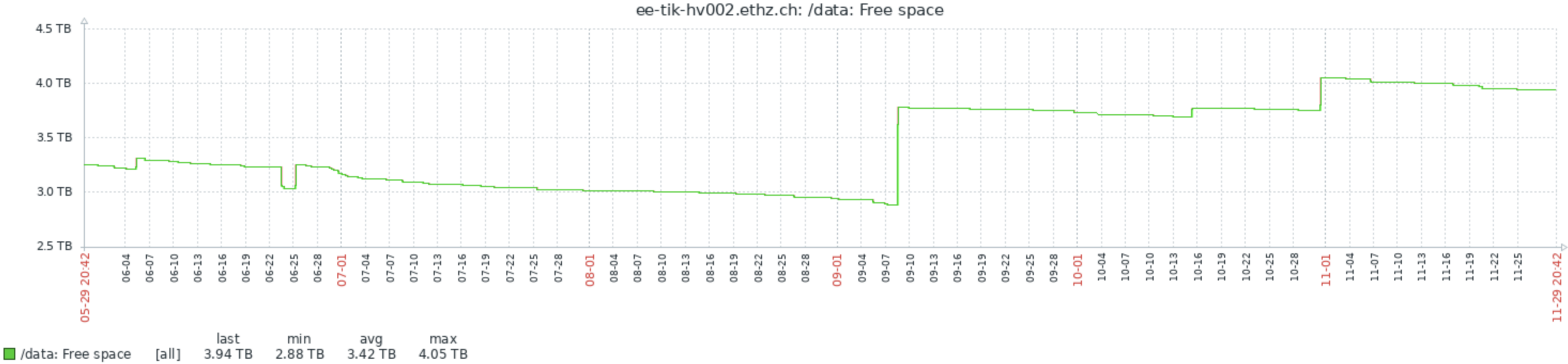
Case Study: Host ee-tik-hv002.ethz.ch

- CPU Utilisation
- Disk Space: Free space
- Load Average
- Memory Usage
- IPMI System Power (Watts)

CPU Utilization

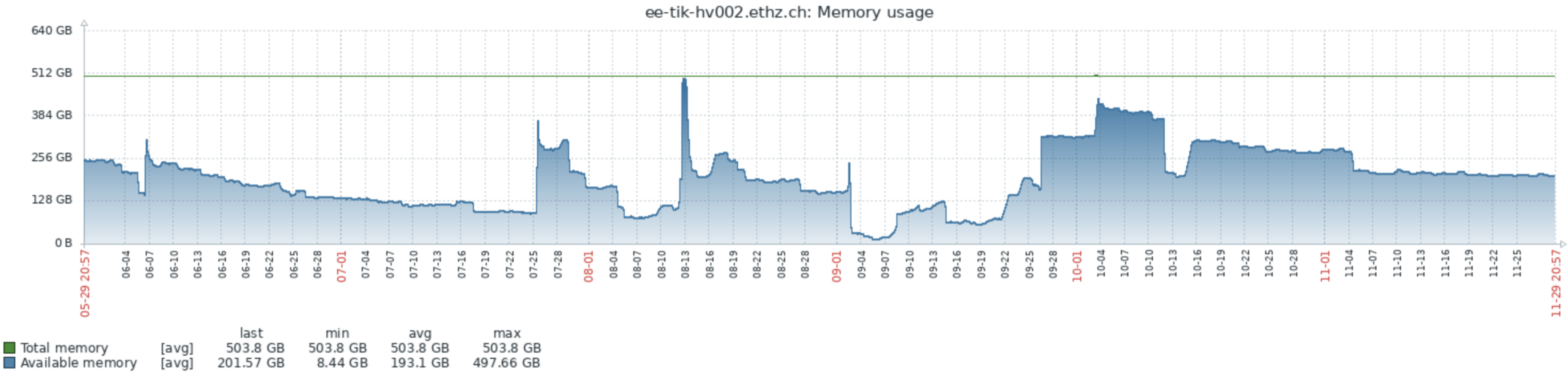


Free Space

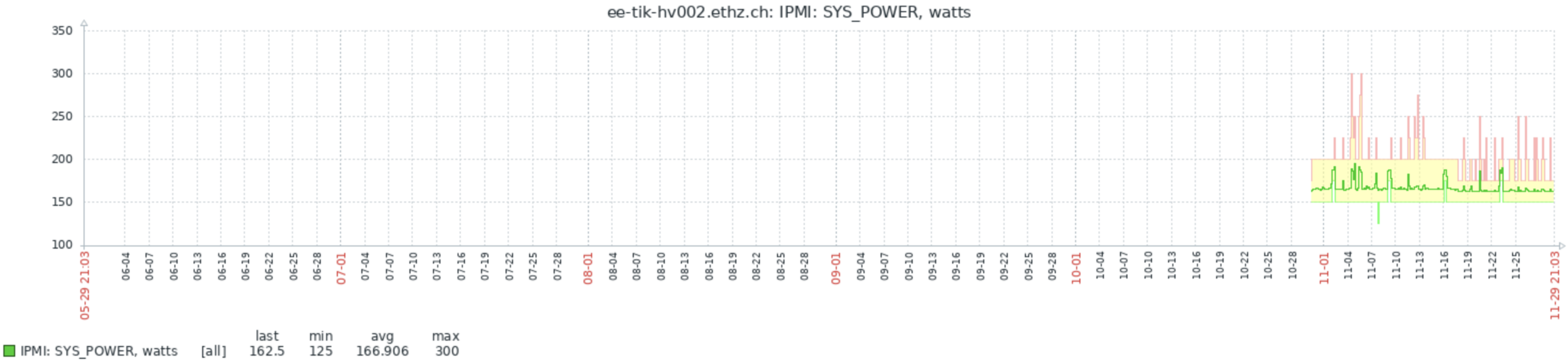


ee-tik-hv002.ethz.ch: /data: Free space
4.5 TB
4.0 TB
3.5 TB
3.0 TB
2.5 TB
05-29 20:42
07-01
08-01
09-01
10-01
11-01
11-29 20:42
last min avg max
/data: Free space [all] 3.94 TB 2.88 TB 3.42 TB 4.05 TB

Memory Usage



SYS_POWER, Watts



Developed Power Measurement Solution

Design Concepts & Requirements

1. **Adapt to Environment:**
   - Fixed Components, Existing Software Stack
2. **Reduce Dependencies:**
   - Programming Language, Libraries, Executables
3. **Simplify Code:**
   - More Code = Higher Maintainment Costs
   - Higher Risk of Human Error
4. **Keep Software Compatibility**

Implementation Diagram

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ANSIBLE

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Server

Server

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Server

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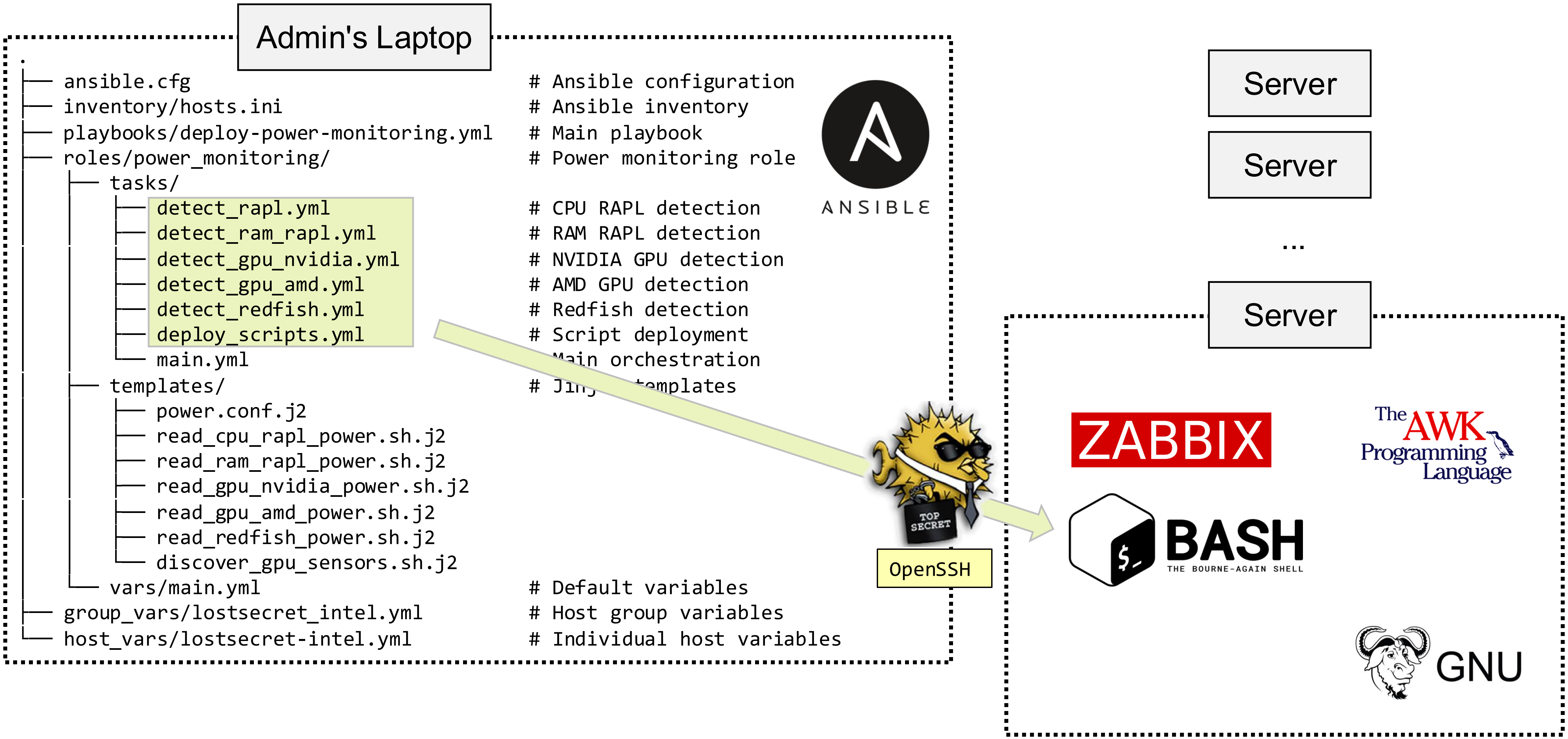

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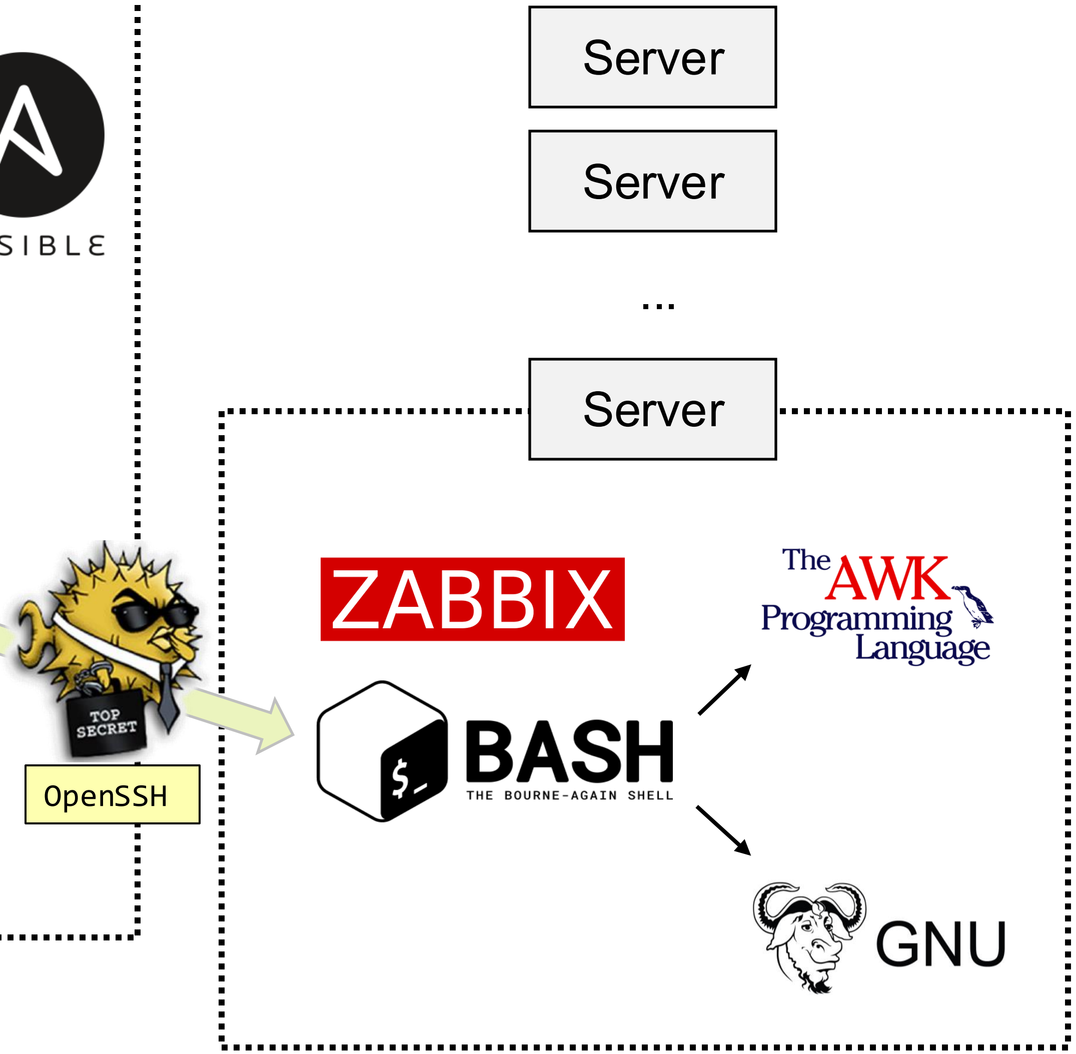

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ANSIBLE

Server

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Server

ZABBIX

The AWK Programming Language

BASH
THE BOURNE-AGAIN SHELL

GNU

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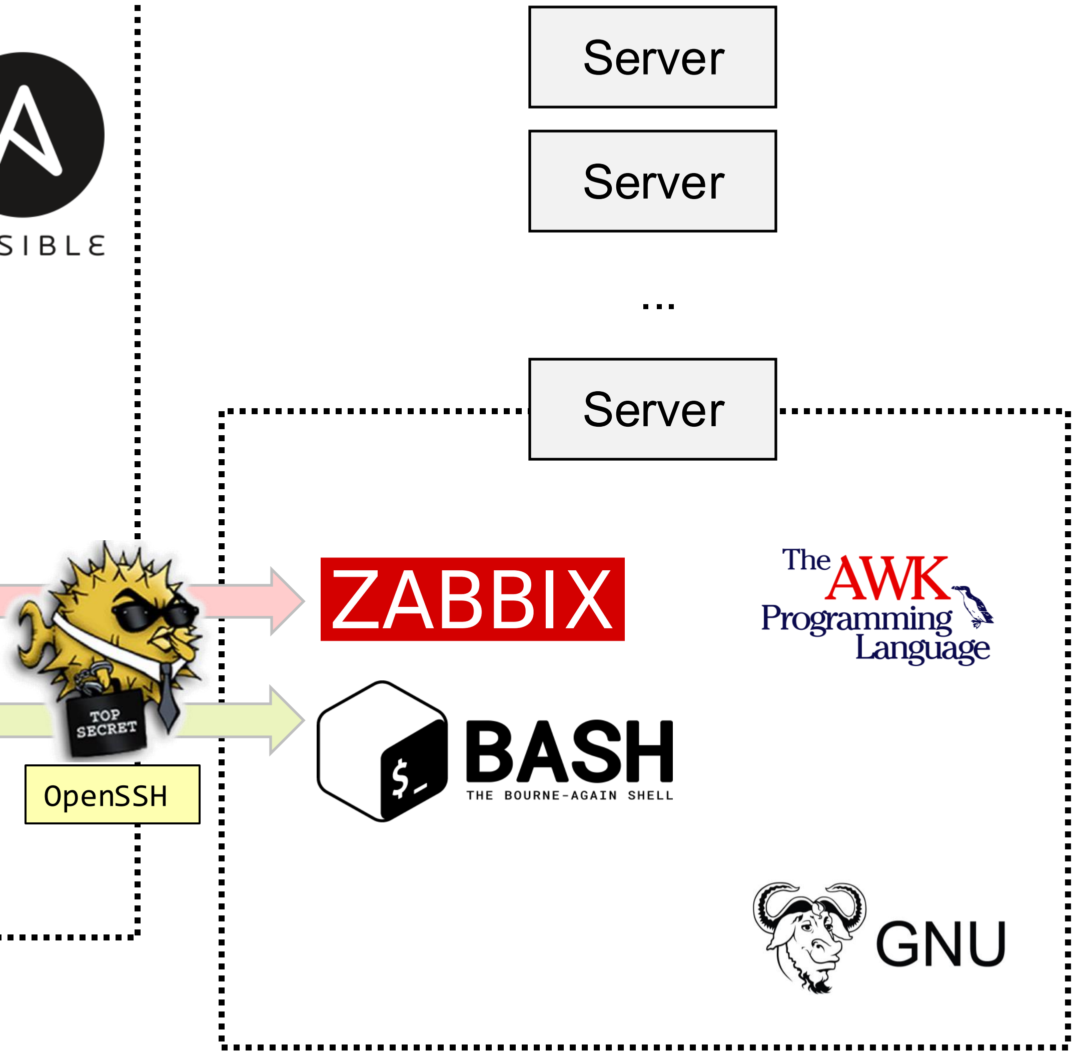
ANSIBLE

Jinja



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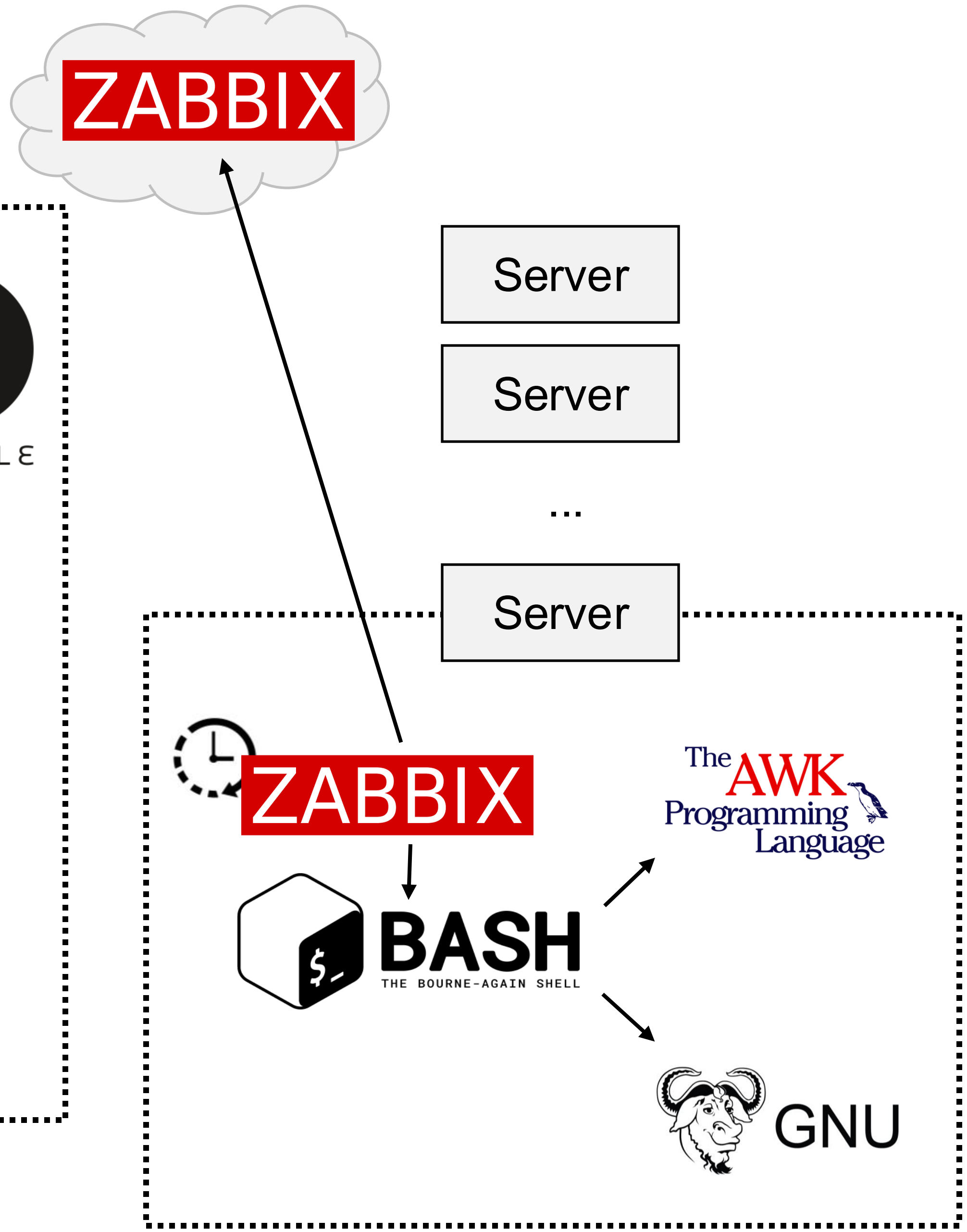

Source Code

https://github.com/ekartashov/ansible-itet-green

Component Analysis

Formulation Component Analysis

- **Goal:** find specific components responsible for majority of energy consumption
- **Databases:** Zabbix, ETH inventory
- No data about actual power usage

Power Supply Unit (PSU):

- Conversion efficiency from AC to DC
- Can waste up to 20% of input power as heat

Central Processing Unit (CPU):

- Consumes 35-60% of total system power
- Newer models: fewer servers needed for workload & each server requires less power for same performance

Analysis PSU

- No data provided → inferred from server chassis models

Idle Power:

- How much power a PSU consumes while non-active
- Lower idle power → more energy efficient server system
- PSU inefficient at lower loads

Full-Load Power:

- Maximum constant power output
- PSU most efficient at 40-60% load
- Lower full-load →more energy efficient

Results PSU

Table 6.1: Typical PSU capacities and power usage for systems in the inventory.

Physical Server Model	**Typical PSU**	**Idle Power (W)**	**Full Load (W)**
Intel S2600 series[8]	2x 750 W Platinum hot-swap	85-120	250-400
Gigabyte R-series (EPYC/Xeon)[9]	2x 1200 W / 1600 W Titanium	110-180	350-700
Gigabyte G482 (GPU server)[10]	2x 2200 W Titanium	250-350	1200-2000
Intel M50CYP series[11]	2x 1300 W Platinum	70-110	220-350
Intel S4600LH[12]	2x 750 W Platinum	70-100	200-300
Desktops (OptiPlex / EliteDesk / ASUS)[13][14][15]	Internal ATX or proprietary PSUs (e.g., Dell OptiPlex 7060, ASUS B365M-K, HP EliteDesk 800 G3)	15-40	80-200

high-performance, heavy compute workloads

least energy efficient (overall)

most energy efficient (server)

most energy efficient (overall)

Replacement Options PSU

Table 6.2: Typical replacement PSU capacities and power usage for systems in the inventory.

	Replacement PSU	Typical Efficiency	Est. Idle Power (W)	Est. Full Load (W)
servers	Supermicro PWS-1K22-1R 1200 W (redundant)[16]	≈88-92%	≈70-100	≈300-450
	Supermicro PWS-1K22A-1R 1200 W (Titanium)[17]	≈90-94%	≈65-95	≈280-430
desktops	be quiet! Straight Power 12 1200 W[18]	up to 93.7% at 50% load	≈50-80	≈250-400
	SilverStone SST-ST1200-PT 1200 W[19]	≈90-94%	≈55-85	≈260-420

➔ lower heat ➔ lower operating cost & better perf/W
➔ Idle Power drops by 10-30 W

Anaylsis CPU

- Explicit data provided
- Additional research (e.g. manufacturer data sheets)

- **Idle Power:**

➢ Energy CPU consumes while not actively used

➢ CPU in idle state majority of time

➢ Lower idle power → better energy efficiency

- **Full-Load Power:**

➢ Actual electrical power drawn when fully utilised

➢ Higher full load → higher energy consumption

- **Thermal Design Power (TDP):**

➢ Max. sustained heat endured by cooling system

➢ Lower TDP → better energy efficiency (less cooling required)

➢ Modern high-TDP CPU more efficient than older low-TDP CPU (sever architecture)

Results CPU

Table 6.3: Typical CPU capacities and power usage for systems in the inventory.

Model Group	TDP Range (W)	Idle System (W)	Full Load (W)
AMD EPYC 7002/7003 Servers (Rome)[20][21]	120-225	25-40	180-270
Intel Xeon S2600 Family[22]	205	20-40	150-220
Intel Xeon E5/ E5-4600 (legacy)[12]	85-155	10-25	90-150

least energy efficient

most energy efficient

Replacement Options CPU

- better perf/W
- performance maintained under 24/7 load
- consolidate workloads

Table 6.4: Typical replacement CPU capacities and power usage for systems in the inventory.

Replacement CPU	Est. TDP Range (W)	Est. Idle System (W)	Est. Full Load (W)
Intel Xeon Silver/Gold 3rd Gen (Ice Lake)[23]	≈150-165	≈20-45	≈120-180
Intel Xeon 3rd Gen (3rd gen Scalable)[24]	up to 270	25-45	180-270
AMD EPYC 9004 (Genoa)[25]	200-400	30-60	200-400
Desktop: Intel Alder Lake/Raptor Lake (12th/13th gen)	35-125	5-20	40-150
Desktop: AMD Ryzen 5000/7000 series (Zen 3/4)[26]	65-105	5-20	50-150
Desktop Core i5/i7 (OptiPlex / EliteDesk / consumer boards)[27]	35-125	5-25	40-150

Conclusion

What We Did?

1. **Gathered All Available Data**
2. **Conducted a Broad Initial Data & Component Analysis**
3. **Evaluated the Power Readings Data Absence Problem Qualitatively**
4. **Formulated Clear Development Guidelines**
5. **Developed and Tested a PoC Power Measurement Solution**
6. **Conducted a Second In-Depth Component Analysis**
7. **Concluded the Limits of What Could Be Done With Initial Data**
8. **Projected What Can Be Done With Actual Power Readings**

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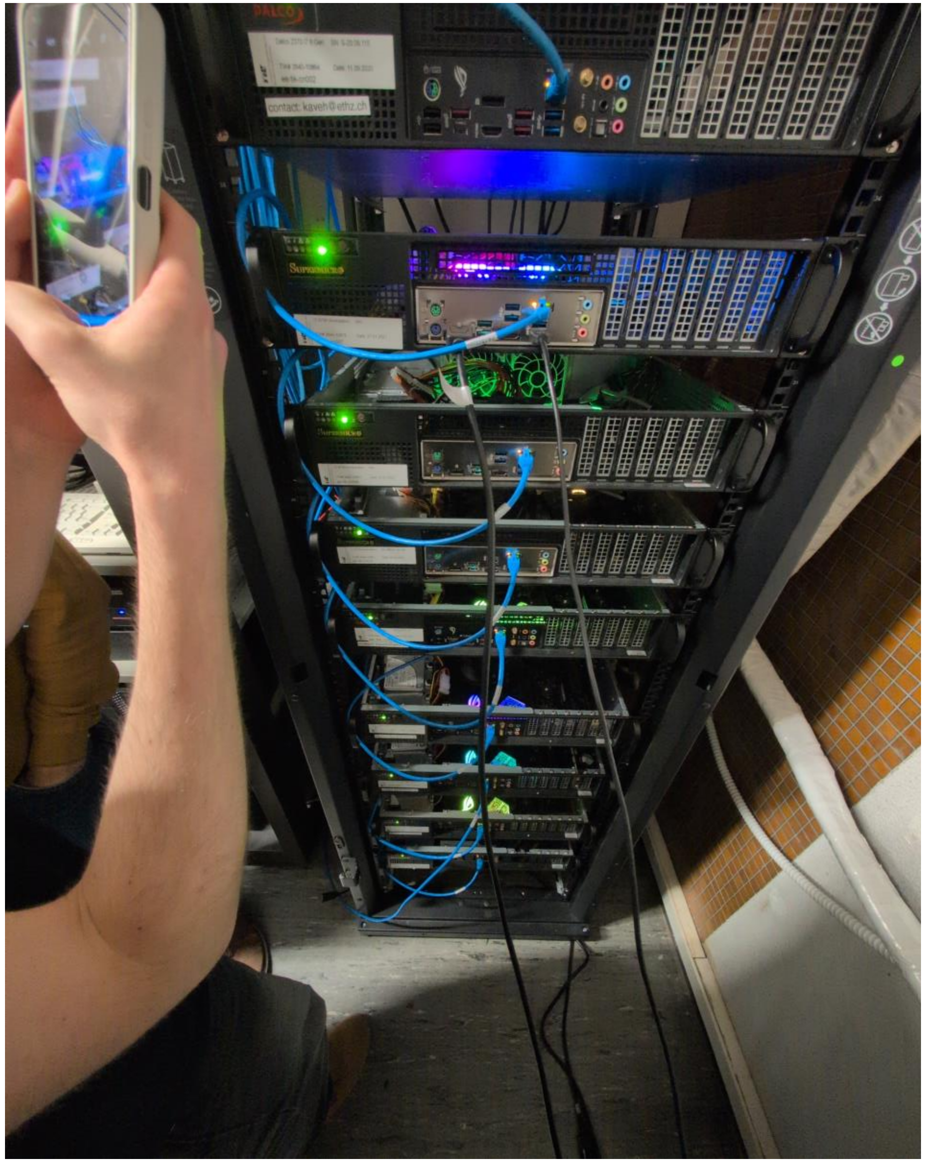

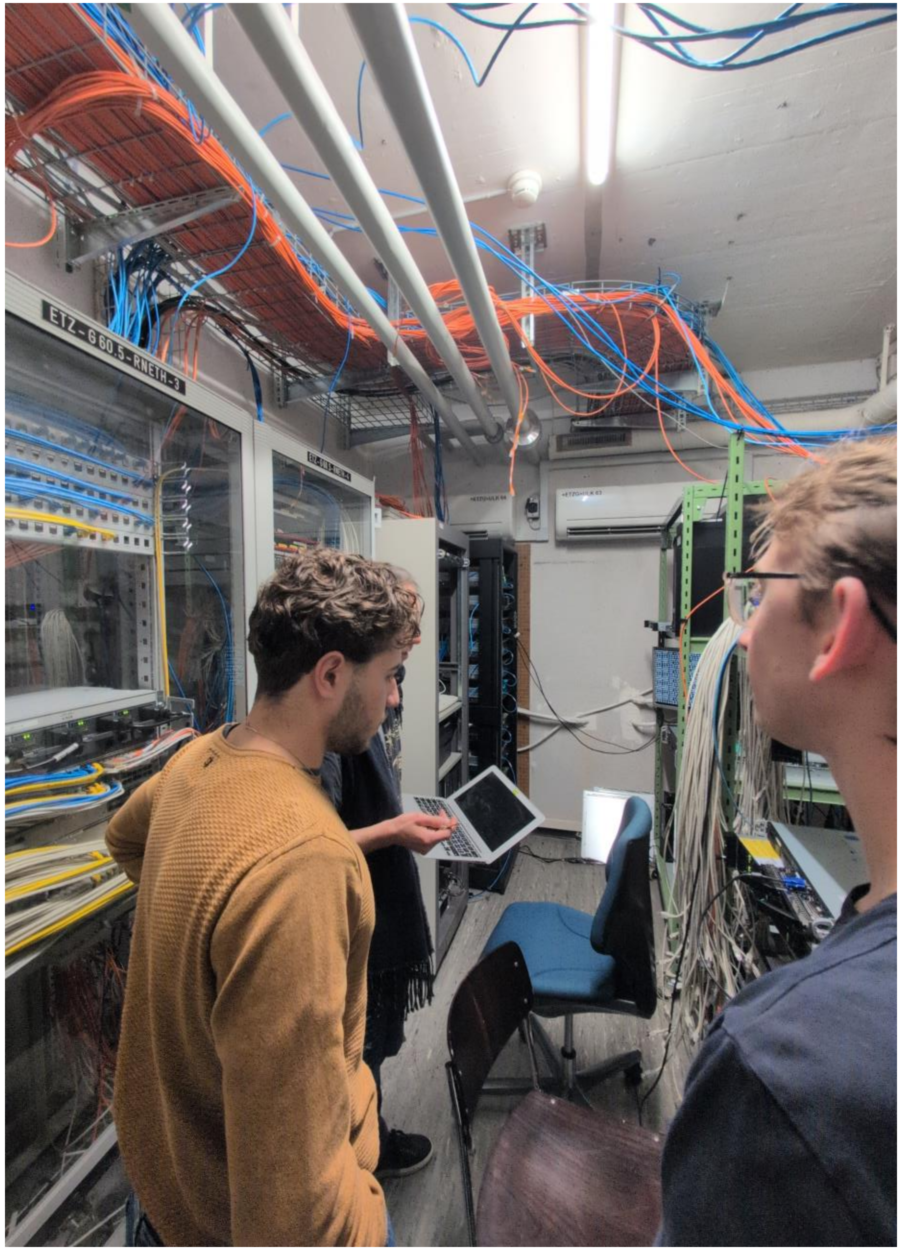

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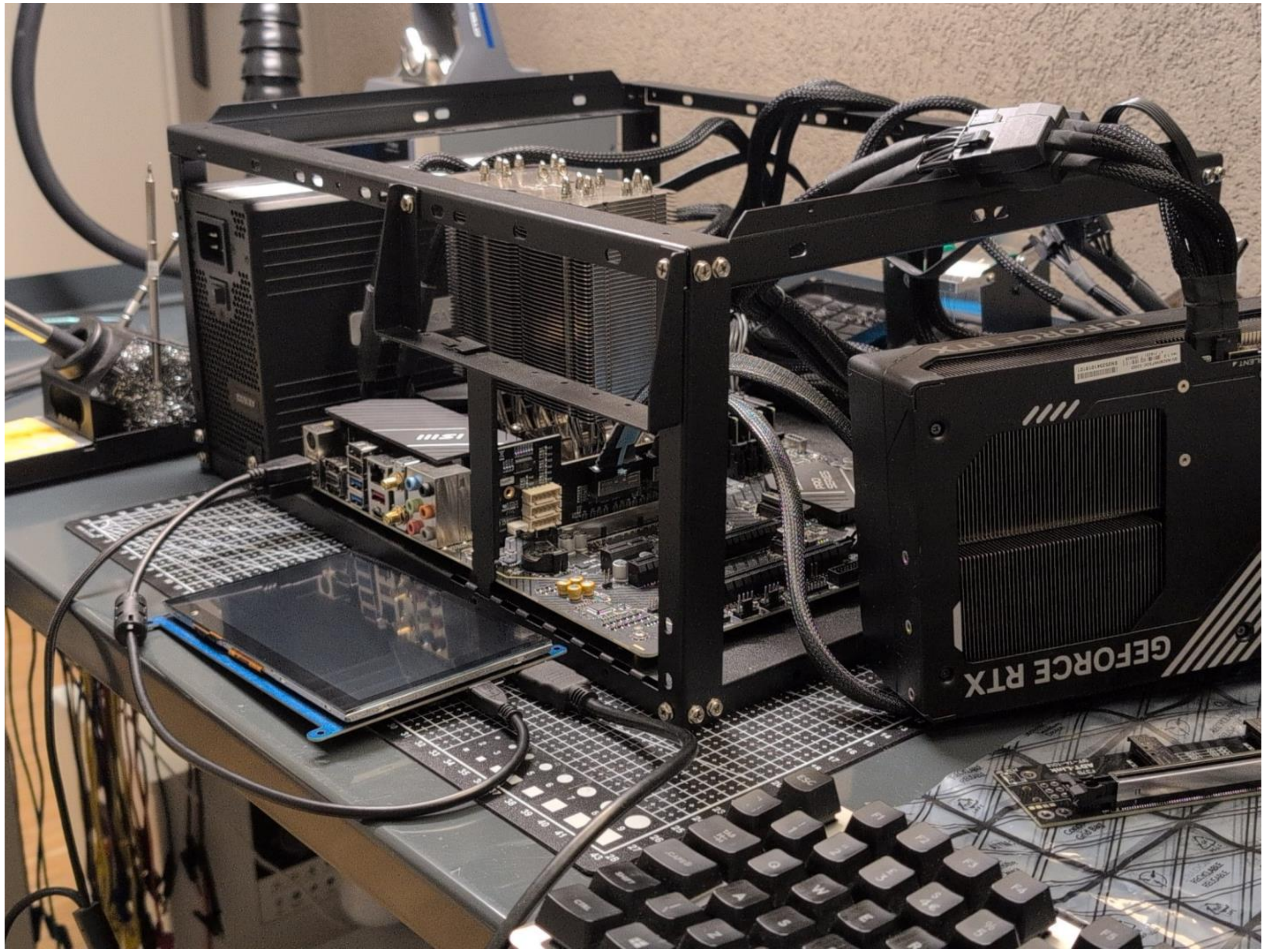
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